

Counterfactual Latent Representations: A Neurosymbolic Approach

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Counterfactual fairness, i.e., asking how a prediction would change under interventions on sensitive attributes, is hard to operationalize for unstructured inputs such as text. We propose a modular neurosymbolic framework that decouples representation learning from causal intervention: a pretrained encoder maps text to a latent representation, a semantic decoder grounds part of this space in a structured causal subgraph, and a neural manipulator learns to transform the latent representation analogously to a symbolic counterfactual operation on the decoded features, trained under a consistency constraint between the two. The approach localizes causal assumptions in an explicit subgraph, trading end-to-end optimization for auditability.

Keywords: Neurosymbolic AI, Counterfactual Fairness, Bias Mitigation, Representation Learning

Reference Format:

Leonhard Kestel and Christoph Kern. 2026. Counterfactual Latent Representations: A Neurosymbolic Approach. In *Proceedings of Fifth European Conference on Algorithmic Fairness (ECAF'26)*. Proceedings of Machine Learning Research, 6 pages.

1 Introduction

Sensitive attributes such as gender [3] and race [4] continue to influence screening and hiring decisions, which is why such tasks are regulated under EU anti-discrimination law [22]. While algorithmic hiring offers an opportunity to reduce human stereotyping and discrimination, algorithms often inherit societal biases from their training data. As a result, fairness concerns in algorithmic hiring are actively discussed by a growing body of research [9].

From a counterfactual fairness perspective, the central question in this dilemma is: “How would a person’s CV look like if its sensitive attribute(s) were different? Would this person be treated differently?” Answering this question requires causal reasoning to generate this person’s data in a counterfactual world, which is a non-trivial task, particularly with unstructured high-dimensional data, such as text.

In this work, we propose a methodology to generate a counterfactual latent representation of a text input. We use a pre-trained variational autoencoder (VAE) to map text to a latent representation, and train a neural manipulator model to transform these representations analogously to a causal intervention (abduction-action-prediction) on observable data w.r.t. a sensitive attribute. This manipulator is supervised by a semantic decoder that maps a fraction of the latent space to a structured representation, thus providing a consistency signal during training. We sketch the architecture, propose a training objective for the manipulator, and discuss its limitations (see Figure 1a).

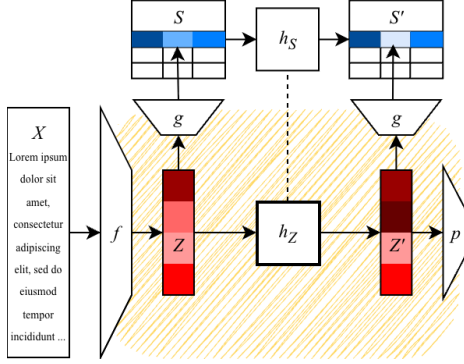
Our work adds to the emerging research at the intersection of neurosymbolic AI and algorithmic fairness [e.g., 1, 10, 21] and illustrates the potential of neurosymbolic approaches for flexible bias mitigation.

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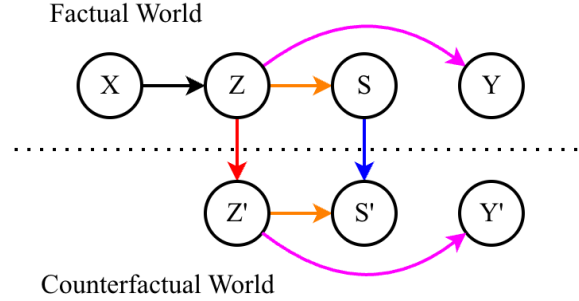
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ECAF’26, September 02–September 04, 2026, Ghent, BE

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(a) General architecture demonstration: an encoder f encodes text data as a representation Z in “latent space” (shaded). A decoder g extracts some relevant structured features S from Z . We define a symbolic manipulation function h_S that transforms S to S' , e.g., by calculating the counterfactual. Knowing S' , we use g as a consistency bridge to train an analogue manipulation function h_Z in latent space.



(b) Graphical illustration of the procedure: Phase 1) A semantic decoder g (orange) is trained to retrieve some structured features S from Z . Phase 2) A counterfactual function h_S (blue) is derived from an SCM to transform S to S' . An analogue latent function h_Z (red) is trained to transform Z to Z' , such that $g(Z') = S'$. A prediction model p (pink) then predicts a target variable Y (or Y') from Z (or Z').

2 Background and Related Work

Counterfactual Fairness. Counterfactual fairness [12] requires that a prediction remains stable under interventions on sensitive attributes, e.g., the predicted job fitness should not change if an applicant had a different gender. Generating such counterfactuals requires a causal model of the data to estimate the effect of an intervention. Path-specific counterfactual fairness [6] relaxes this by intervening only on selected pathways.

Fair representation learning. Following *Learned Fair Representations* [23], the fairness research community has turned its gaze onto representation learning. Many approaches have thus been developed to implement a constrained encoding of data to a lower dimensional representation via adversarial learning [e.g., 14], dimensionality reduction [11, 18] or with variational autoencoders [e.g., 7, 13, 16]. A recent approach leveraged VAEs to create latent representations of text with an independence constraint [15]. Similarly to other literature, they train an encoder to directly map an input X to a debiased representation Z^* . Closer to ours, Creager et al. [7], Oh et al. [16] also adopt a modular design that separates encoding from debiasing.

Deep counterfactual inference. Deep Structural Causal Models [17] operationalize Pearl’s abduction-action-prediction procedure using normalizing flows and VAEs as (pseudo-)invertible connectors, with extensions to high-resolution images [19] and graph-structured data [20], the latter applied to a counterfactual fairness setting. Unlike these works, we do not use the VAE as a functional connector within an SCM but only to encode data into a low-dimensional space, where a learned manipulator approximates the intervention.

Neurosymbolic latent reasoning. The idea for this work stems from a neurosymbolic planning approach called *LatPlan* [2], which divides the solution of image puzzles in two steps: neural encoding of the image to a discrete latent representation and symbolic latent space reasoning. We adopt this modular approach, although we do

not enforce discrete latents, but train a continuous latent manipulator approximating a symbolic procedure, i.e. calculating a counterfactual.

Our contribution sits at the intersection of these threads: a modular, neurosymbolic architecture that approximates counterfactual interventions in the latent space of a VAE.

3 Problem Formulation

Let X denote a text input (e.g., a CV) and A a sensitive attribute (e.g., gender), and let Z be a latent representation produced by a pre-trained encoder. We assume access to a structural causal model M over a set of observable variables S that captures the causal effect of A on relevant features of X (e.g., the style in which skills are presented). Given a counterfactual value a' for A , our goal is to learn a latent manipulator that produces $Z'_{a'}$, the latent representation X would have had under $do(A = a')$. A downstream predictor p then satisfies counterfactual fairness if $p(Z) = p(Z'_{a'})$ for all a' .

4 Procedure

We propose a modular and interpretable approach using a pre-trained encoder mapping X to Z and a neurosymbolic manipulator altering Z to Z' . These are part of a 2-phase training framework to obtain a modular model for counterfactual inference on text data.

As basis, we use the pre-trained encoder, such as *LangVAE* [5], as a function $f : X \rightarrow Z$ to obtain the latent representation of text data.

Phase 1: Semantic Decoder. We train a semantic decoder function $g : Z \rightarrow S$ that learns a mapping from the VAE's latent space Z to a structured tabular representation of selected features S , including the sensitive attribute A , involved the SCM M . M does not necessarily model the entire data generating process of X , but merely represents the subset of causal paths, where A has an unwanted spurious influence. Hence, the semantic decoder only decodes a fraction of Z .

Phase 2: Neurosymbolic Manipulator. Given M , we can derive a function $h_S : S \rightarrow S$ that calculates the counterfactual S' to S w.r.t. the sensitive attribute A via abduction-action-prediction on M . A neural manipulator model $h_Z : Z \rightarrow Z$ then learns the transformed latent representation $Z' := h_Z(Z)$ analogously to $h_S(s)$. It thereby aims to satisfy the consistency constraint (see Fig. 1b):

$$h_S(g(z)) = g(h_Z(z)) \quad (1)$$

Note that the parameters of the decoder g are frozen. h_Z can be trained by minimizing the following objective, in which the first additive term represents the consistency constraint given above using a distance metric d_S appropriate for the structured data space S , while the L_1 regularizer promotes sparsity and the L_2 term penalizes strong deviations from the original representation Z :

$$\mathcal{L} = \underbrace{d_S(h_S(g(Z)), g(h_Z(Z)))}_{\text{Consistency Constraint}} + \underbrace{\alpha \cdot \|h_Z(Z) - Z\|_1}_{L_1: \text{Sparsity Penalty}} + \underbrace{\beta \cdot \|h_Z(Z) - Z\|_2^2}_{L_2: \text{Deviation Penalty}} \quad (2)$$

This loss function ensures that h_Z changes Z as much as it needs to be consistent ($h_Z \sim h_S$), but as little and selectively as possible. α and β are hyperparameters.

Downstream Task. Given the trained manipulator h_Z , we are able to get a factual latent Z and its (path-specific) counterfactual Z' from given text X . We can then use these for various downstream tasks, such as bias mitigation, e.g., by penalizing a prediction model p if the prediction $p(Z)$ differs from the counterfactual prediction $p(Z')$.

5 Data

Both the decoder training and the manipulator’s empirical evaluation are ongoing. We will conduct experiments on a synthetic and a real world dataset.

5.1 Synthetic Data

The reasons to create a synthetic dataset are twofold: a dataset with matching X and S is needed to train the semantic decoder described in phase 1 of Section 4. Also, we want to create controlled conditions to test our hypothesis that a manipulator model is trainable as a latent analogy to a symbolic operation on structured data. We first construct an SCM modeling some causal paths originating from A and involving S as a set of descendants. We then use it to sample a structured tabular dataset. Finally, we use an LLM to create a text instance (i.e., a CV) from each simulated observation to obtain X .

5.2 Real World Data

To test the applicability of the framework, we implement it with the *Bias in Bios CV* dataset [8], where gender is the sensitive attribute and profession is the target. There are two major challenges regarding this transfer to the real world: first, we need to make strong assumptions about causal relationships involving the sensitive attribute we are targeting to construct an SCM on S . Furthermore, we do not know S , but need to estimate it using a pre-trained semantic decoder.

6 Discussion

We propose a modular framework that separates representation learning from causal intervention: a pretrained text encoder produces a latent representation, and a neural manipulator transforms it analogously to a symbolic counterfactual operation on a structured decoding. This decomposition trades end-to-end debiasing for a more auditable inference pipeline, in which the causal assumptions live in an explicit, inspectable subgraph rather than implicitly inside an encoder.

The framework inherits the standard SCM assumption of no hidden confounders, which is required for abduction and makes constructing a defensible causal subgraph difficult in practice. Noise abduction also requires full observability of S when computing counterfactuals, which is restrictive under missing data. The reliance on an explicit causal graph is by design, as it constrains the manipulator’s behavior to known pathways, but it also caps the framework’s applicability to settings where such a graph can be specified. The central open question is whether the latent manipulator learns a faithful analogue of the symbolic operation in practice, particularly when the semantic decoder is imperfect or trained on synthetic data and applied to real text. We see this as the immediate empirical priority and as a sharp test of whether neurosymbolic decomposition delivers on the flexibility and audibility that motivated the design.

Acknowledgments

We thank Philip Boustani for his input in discussions on causal inference and the foundations of this framework.

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